خدمتكم طريق خضناه لرضى الله

2021

أسئلة سنوات

ميد

إقتصاد هندسي

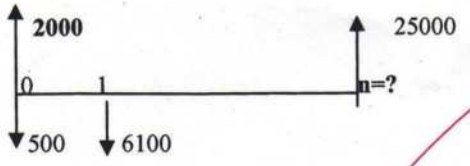


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ENGINEERING ECONOMY first exam

NAME..... الإسم:.... الشعبة:- (٩ – ١٠)

Q1: what is the Equivalent Values (X), (Use ***i.= 15%*** per year). Show the calculation for your answers



$X_o = -2000 + 500 + 6100(P/F\ 15\%, 1)$

$= -1500 + 6100(0.86957)$

$= \boxed{3804.377}$

$F = P(F/P\ 15\%, n)$

$25000 = 3804.377\ (F/P\ 15\%, n)$

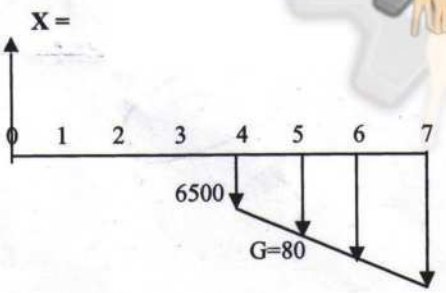
$3804.377 \quad 3804.377$

$(F/P\ 15\%, n) = 6.57$

7.0757
6.1527 → 14
6.57 → n
5.8502 → 13
6.1527

$\frac{n}{1} = \frac{14-n}{14-13}$

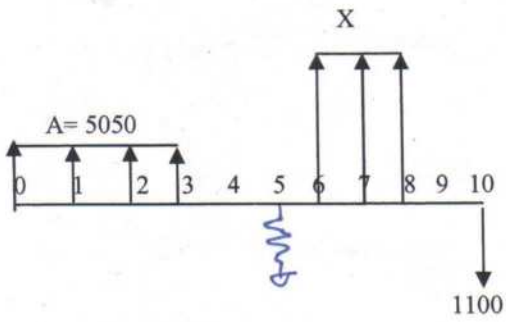
$n = 13.45$ year final Answer



$X_o = 6500(P/A\ 15\%, 4)(P/F\ 15\%, 3) + 80(P/G\ 15\%, 4)(P/F\ 15\%, 3)$

$= 6500(2.8549)(0.65752) + 80(3.7864)(0.65752)$

$= \boxed{12400.67}$

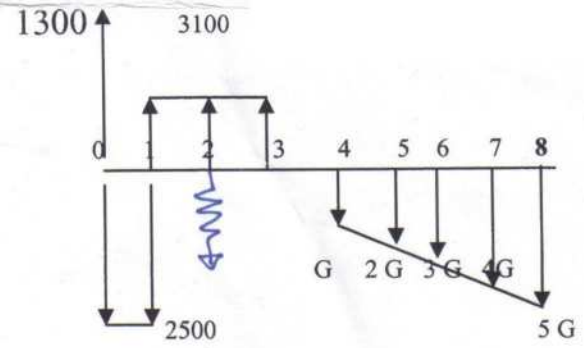


$-5050(F/A\ 15\%,4)(F/P\ 15\%,2) + 1100(P/F\ 15\%,5) = X(P/A\ 15\%,3)$

$-5050(4.9933)(1.3225) + 1100(0.49718) = X(2.2832)$

$X = -14366.45$

اي أن X بسالب.



$-1300(F/P\ 15\%,2) - 3100(P/A\ 15\%,3)(F/P\ 15\%,2) + 2500(F/A\ 15\%,2)(F/P\ 15\%,1) = G(P/G\ 15\%,6)$

$-1300(1.3225) - 3100(2.2832)(1.3225) + 2500(2.1500)(1.1500) = G(7.9367)$

$G = -617.202$

اتجاهها للأسفل

Q2 If R =30%per year compounded continuously find the effective interest rate for quarter compounded monthly

Q4:- Put sign (✓) for the correct answer and sign (✗) for wrong answer

A) (✓)Economics is The study of how limited resources is used to satisfy unlimited human wants

B) (✓)Inflation: Money repaid in the future will "value" less

C) (✓)Good information plus skills gives good data

D) (✓)Principal is The fee that a borrower pays to a lender for the use of his or her money.

E) (✗)factor P (F/P,i%,n) equals $P=F/(1+i)^n$

F) (✗) The Future value is 7346.64 if Present is 5000 invested for 5 years on 8%per year SIMPLE INTEREST RATE

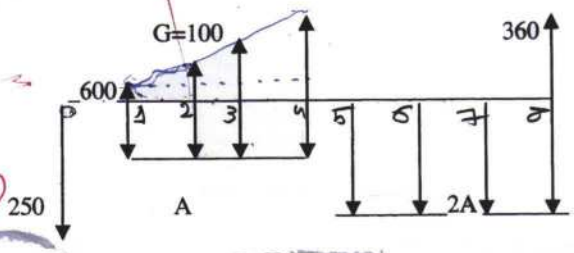
$F = P + P \cdot i \cdot n$

$= 5000 \times \frac{8}{100} \times 5$

ENGINEERING ECONOMY first exam

NAME..... الإسم. الشعبة

Q1: a:- what is the Equivalent Values (X), (Use *i.= 12%* per year). **Show the calculation for your answers**



$A[F/A, 12\%, 4] + 2A[P/A, 12\%, 4] = \{250[F/P, 12\%, 4] - 360[P/F, 12\%, 4] - 600[F/A, 12\% 4] - 100[F/G, 12\%, 4]\}$

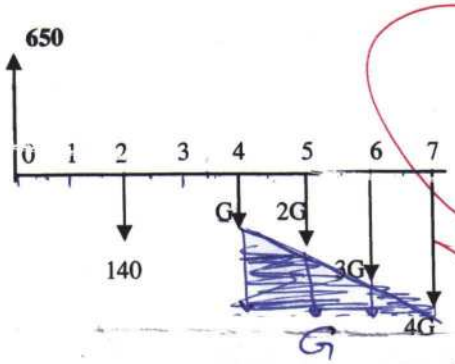
$A[4.7793] + 2A[3.0373] = \{250[1.5735] - 360[0.63552] - 600[4.7793] - 100[6.4944]\}$

$A[10.8539] = 393.375 - 3745.8072$

$A = \frac{-3352.4322}{10.8539} = -308.87$ unit

القيمة المكافئة للـ A عند نهاية السنة 4

Q1: b:-



$G[F/G, 12\%, 5] = [-650[F/P, 12\%, 7] + 140[F/P, 12\%, 5]]$

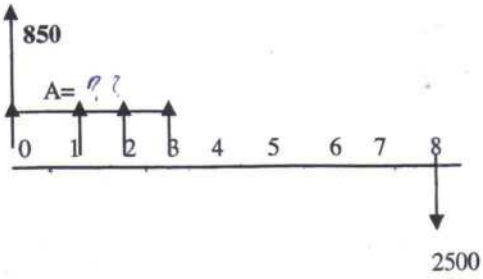
$G[11.273] = \{-650[2.2106] + 140[1.7623]\}$

$G = \frac{-1190.168}{11.273}$

$G = +105.577$ unit

القيمة المكافئة للـ G عند نهاية السنة 7

Q1: c:-



$$-A[F/A, 12\%, 4] = -850[F/P, 12\%, 3] + 2500[P/F, 12\% 5]$$

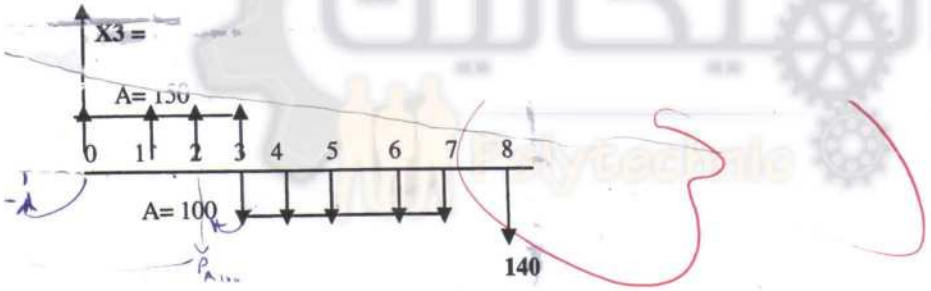
$$-A[4.7793] = -850[1.4049] + 2500[0.56743]$$

$$A = \frac{224.41}{-4.7793} = -46.955 \text{ unit}$$

هذه القيمة الكاشفة لـ A عند نهاية العمر 3

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Q1: d:-



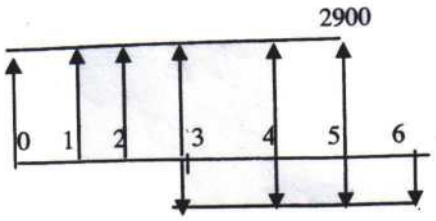
$$X_3 = -150[P/A, 12\%, 4][F/P, 12\%, 1] + 100[P/A, 12\%, 5][P/F\ 12\%$$
$$+140[P/F, 12\%, 8]$$

$$-X_3 = -150[3.0373][1.1200] + 100[3.6047][0.79719] + 140[0.40388]$$

$$X_3 = +166.36 \text{ unit}$$

القيمة الكاشفة لـ X_3 عند الصفر الأفقي.

Q1: e:-



A =

$$A[F/A,12\%,4] = -2900\,[F/A,12\%,6][F/P,12\%,1]$$

$$A[4.7793] = -2900\,[2.1151][1.1200]$$

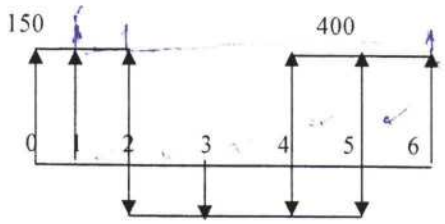
$$A = \frac{-26357.6448}{4.7793} = -5515.00 \text{ unit}$$

ENGINEERING ECONOMY first exam **A**

NAME.....

الشعبة:- ١١-١٢:٣٠ ن ر الإسم:

Q1:- what is the Equivalent Value of (X (Use ***i.= 12%*** per year). Show the equations (CALCULATION)for your answers



X= A =

X= 400.9442

-3

09/20

A_1

F = 150 (F/A) 12% 3

150 * 3.3744

= 506.16 (P/F) 12% 1

= 451.93

A_2

F = 400 (F/A) 12% 3

= 400 * 3.3744

= 1349.76 (P/F) 12% 5

= 765.894

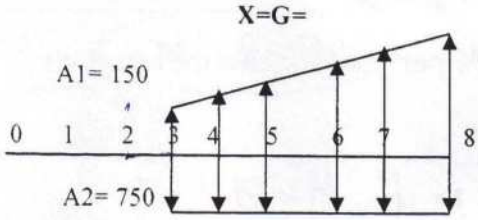
P = 1217.824

A = P (A/P) 12% 4

= 1217.824 * ——

= 400.944

Q2:- what is the Equivalent Value of (X),IF Equivalent Values ofA2 is Equal to the Values of A1+G (Use ***i.= 9%*** per year). Show the equations (CALCULATION)for your answers



X= 400.05

$P = A_2 (P/A_2)9\%\ 6$

$= 750 \times 4.4859$

$P = 3364.425$

$P = 150 (P/A_1)9\%\ 6$

$= 150 \times 4.4854$

$= 672.585$

$G = 4037.31 (G/P)9\%\ 6$

$\frac{1}{10.092}$

Q3:-

a-How long would it take for $200 to accumulate to $1500 at an interest rate of 11% per year?

Answer:

n= 18.7038

-2

$P = F(P/F)11\%,n$

$200 = 1500(P/F)11\%,n \rightarrow 0.13333$

18.7-:

19 → 0.13768

n → 0.13333

20 → 0.12403

19 → 7.2633

n → 7.5

20 → 8.0623

$$\frac{19-n}{19-20} = \frac{0.13768-0.13333}{0.13768-0.12403} = \frac{0.0435}{0.01365} = 0.31868$$

$$\frac{19-n}{19-20} = \frac{7.2633-7.5}{7.2633-8.0623} = \frac{-.2367}{-.799} = +0.2962$$

+19

b- For an interest rate of 15% per year, compounded quarterly, the effective interest per six months(semiannually) is how much? (Equation and the answer)

i/semi= 1.25×10^{-4}

i/s

4/s

$i = (1+\frac{R}{n})^n - 1$

$= (1+\frac{15\%}{4})^4 - 1$

$= 0.15866$

$1+\left(\frac{3.75}{3}\right)^3 - 1$

$= 0.0125 \times 2$

$= 0.025$

$i_{s} = (1+\frac{0.025}{2})^2 - 1 \quad 1.25 \times 10^{-4}$

ENGINEERING ECONOMY first exam **A.**

NAME.....

الشعبة:- ١١,١٥ - ١٢,١٥

1-What value of annual(A) uniform payments over years 1 – 5 is equivalent to a payment schedule of 200, 200, 240, 240, 240 for years 1-5 respectively at an interest rate of 10%?(Factor notation only)

الحل على ورقة ١ اجابة

A = 277.62

2- About how long would it take for $1000 to accumulate to $8000 at an interest rate of 9% per year? (Factor notation only)

n = 24.1297 year

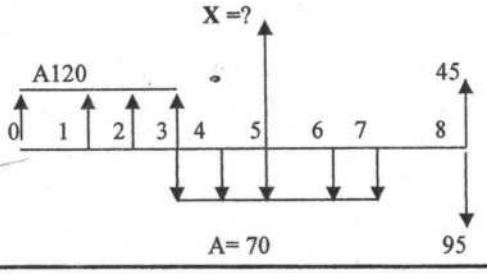
3- If you well invest $50,000 each year for four years starting 2 years from now. What is the present worth of the investment at an interest rate of 10% per year? (Factor notation only)

P = 130570

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4-For an interest rate of 12% per year, compounded quarterly, the effective interest per six months is how much? (Equation only)

5- What is the Equivalent Value of **X** in year 5 of all the cash flow diagram showing down interest rate of 12%



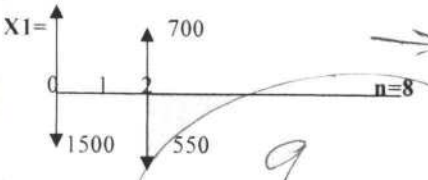
6-a. what is the interest?

b .why interest rate?

ENGINEERING ECONOMY first exam **B.**

NAME..... الإسم: 8_٣,٩ (ن /) الشعبة:-

Q1: what is the Equivalent Values (X), (Use ***i.= 8%*** per year). Show the calculation for your answers



a:-1371,399 b:-2538,33 c:-3014,37 d:- non of all

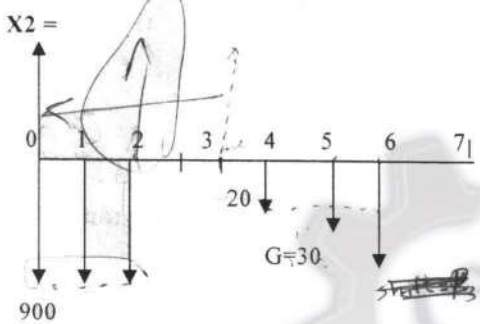
$P_{700} = 700(0.85734) = 600,1 \downarrow \Leftarrow P = A(P/A, 8\%, 2)$

$P_{550} = 550(0.85734) = 471,5 \uparrow \Leftarrow P = A(P/A, 8\%, 2)$

$X = 471,5 - 600,1 - 1500 = (1628,6 \downarrow)$

$P = 150(P/F\ 8\%\ 2)$

$128,601$



a:-4168,044 b:-2577,765 c:-4417,84 d:-non of all

$P_A = 20(2,5771) = 51,542 \uparrow \Leftarrow P = A(P/A, 8\%, 3)$

$P_G = 30(2,445) = 73,351 \Leftarrow P = G(P/G\ 8\%, 3)$

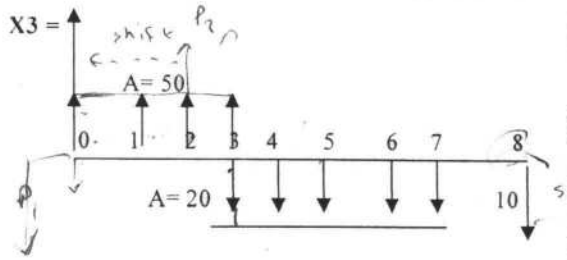
$P_3 = 51,542 + 73,35 = 124,892 \Leftarrow P_3 = P_A + P_G$

$shift\ P_3 - P_0 = 124,892(0,73503) = (91,749 \uparrow)$

$P_{900} = 900(2,5771) = 2319,39 \uparrow \Leftarrow P = A(P/A, 8\%, 3)$

$P = 900(P/A\ 8\%\ 3) + 30(P/G$

$X_2 = P_3 + P_{900} = 2411,18$



a:--194,32 b:--104,985 c:-447,758 d:- non of all

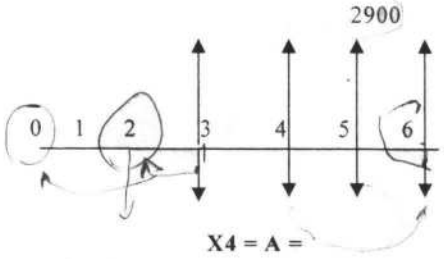
$P_1 = 50(3,3121) = -165,605 \downarrow \Leftarrow P = A(P/A\ 8\%, 4)$

$P_2 = 20(3,9927) = 79,854 \uparrow \Leftarrow P = A(P/A\ 8\%, 5)$

$shift = P_2 = (79,854)(0,85734) = 68,46$

$P_3 = 10(0,54027) = +5,4027$

$X = P_1 + P_2 + P_3 = -165,605 + 68,46 + 5,4027 = (-91,745)$



a:-3131,980 b:-2302,086 c:-2899,968 d:- non of all

$P = 2900(3,3121) = 9605,96 \downarrow \quad P = A(P/A\ 8\%, 4)$

$A = 9605,96(0,30192) = 2900,2 \uparrow \quad A = P(A/P\ 8\%, 4)$

Engineering Economy		Al-Balqa` Applied Universtiy
First Exam A Time: 60minutes		Faculty of Engineering Technology
Name: ID:		Department of Civil Engineering

Give the final answer for the following questions (show your calculations)

Q1: The amount of an interest earned on 3000 for 6 years at 10%/yr simple interest is---1800---

Q2: If a sum of JD 3000 is deposited now, JD 9000 three years from now, and JD 3000/year in years 6 through 10, the amount in year 10 at i=20% is---73145.1

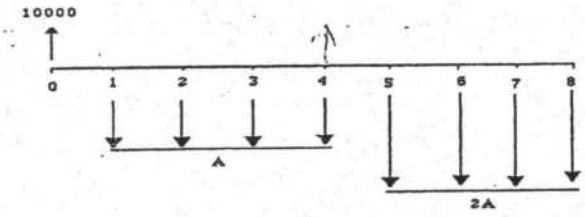
Q3: If JD 8000 is invested now that is expected to give a JD18000 after 5 years, the interest rate for this investment is ---35%---

Q4: How long does it take for an amount of money to be increased seven times the initial amount at an interest rate of 15 % yearly? n= ---14---

Q5: If a person deposited JD 450 now and JD 400/year for 7 years starting at year 1 then his withdrawal at the end of the 10th year will be---6185.78--- i = 10%

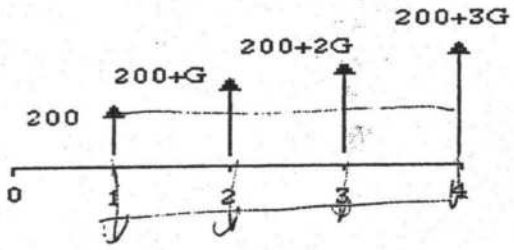
Q6: How much money would you have to pay each year in seven equal payments, starting three years from today, to repay a 20,000 JD loan at i=15%?---6389.3---

Q7: For the cash flow shown, the value of A when i=15% is---1633.9---



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Q8: For the cash flow shown below, the value of A disbursements that will make an equivalent increasing arithmetic gradient with G=100JD at an interest rate of 20% per year is---327.43---



Q9: A project is to be constructed with an initial cost paid as follows: JD 20,000 now and five equal payments each of 4000 started 4 years from now. The maintenance cost is 1500 for the first year that is increased 300 yearly thereafter. An extra cost of 2500 is needed every 4 years. The expected income is 9500 yearly started 5 years from now, and the salvage value is 5000 after 12 years, Find the present value of the project (use i=5%) PW=---23735---

present worth

With Our Best Wishes

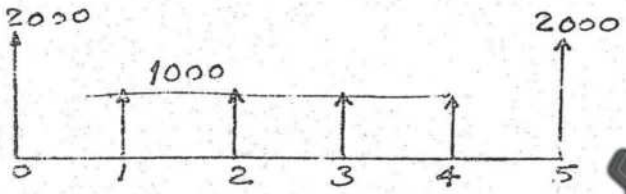
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.com/muslimEngineerPage

Q4: Select the correct answer for the followings (Show your calculations)

1. If the interest rate is 7% per semiannual compounded monthly, the effective interest rate per quarter is:
   a) 3.85% b) 3.54% c) 3.5% d) 3.67%

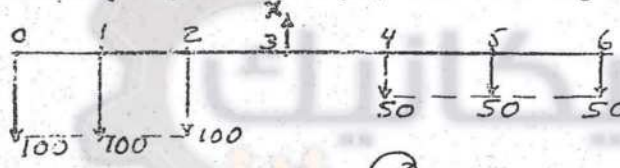
2. If the interest rate is 16% per year compounded continuously, the effective interest rate Semiannually is :
   a) 8.33% b) 17.35% c) 8.16% d) 8.85%

3. For the shown cash flow ,which of the following Statements is incorrect i=15%



a)1000(P/A,15%,4)+2000+ 2000(P/F,15%,5)
b)1000(P/F,15%,5)+1000(P/A,15%,5)+2000
c)[1000(F/A,15%,5)+1000](P/F,15%,5) +2000
d) {1000(F/A,15%,4)+2000}(P/F,15%,4) +2000

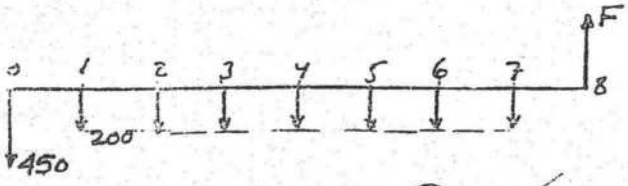
4. If i=10%per year ,the value of (X) in the following cash flow is



a) 450 b)462 c) 457 d) 483

5. The amount of interest earned on 2000 for 5years at 10% simple interest per year is
   a) 3000 b)3221 c) 1000 d) 1221

6. If i=10%per year ,the value of (F) in the following cash flow is



a)2550 b)2862 c) 3052 d)3252

Q5: A factory is to be constructed by an initial cost paid as follows: JD20000 now and five equal payments each of 4000 started 4 years from now. The A.O.C is 1500 for the first year increased 300 yearly thereafter. An extra cost of 2500 is needed every 4 years. The expected income is 9500yearly started after 5 years from now, the salvage value is 5000 after 12 years. If the interest rate is 10% per year :

1-Draw the cash flow diagram.

Find the present value of the project.

With our best wishes

Q2: Circle "T" (for True) or "F" (for False) for each of the following statements.

A) The change in the amount of money over a given time period is called the interest rate ~~T~~ F

B) To invest any amount of money for one year at a (6%) compound interest rate, is better than at a (6%) simple interest rate. ~~T~~ F

C) The factor (P/F,i%,n) equals $P=F/(1+i)^n$ T F

D) The Good Decision is depend on Good Information T F

Q 3:**a:-** calculate the future value if you invest 8500amount of money for 8years at 7%simple interest rate

$I = R \times i = 8500 \times 7\% = 595$

$i = 595 \times 7 = 4165$

$4165 + 8500 = 12665$

$F = P + I$

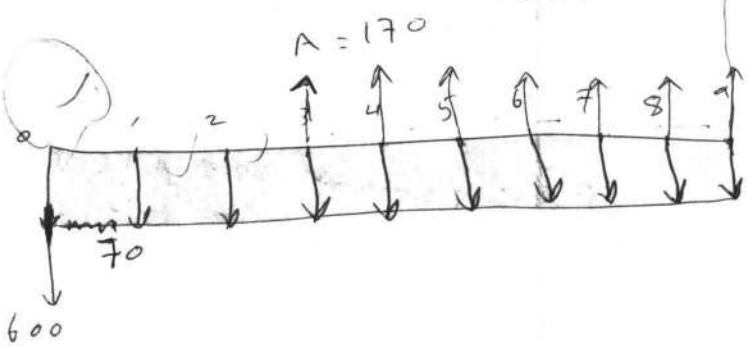
$P + Pni$

$1 + 2 \quad 60$ الجواب

b:- Draw and calculate the equivalent value at zero (0)time the for a project with the following payments:

- first cost = JD 600;
- operating cost = JD 70,
- Annual Income for the last 6 years = JD 170;
- Salvage Value = 200;

if n, = 9 years .i=13%compounded interest $i = 13\%$



$P_A = 70 \times (5.1316) = 359,212$

$P_{170} = 170 \times (3.9975) = 679.57$ shift (0.3075) = 470

$P_{200} = 200 (0.33288) = 66,576$

$= -177.364 - 600 = -777,364$

$422,636$

GOOD LUCK

Q2: Clearly circle "T" (for True) or "F" (for False) for each of the following statements.

a) To invest JD(1000) for 3 years at a (6%) compound interest rate, is better than at a (7%) simple interest rate. T (F)

(CALCULATIONS)

Comp $F = P(1+i)^N = 1000 \times (1+6\%)^3 = 1191.016$

Simp $F = P + Pin = 1000 + (1000 \times 7\% \times 3) = 1210$

That time is $F.(P/F, i\%, 1)$

b) The (P/A,i%,N) factor equals (N * (P/F,i%,1)) T (F)

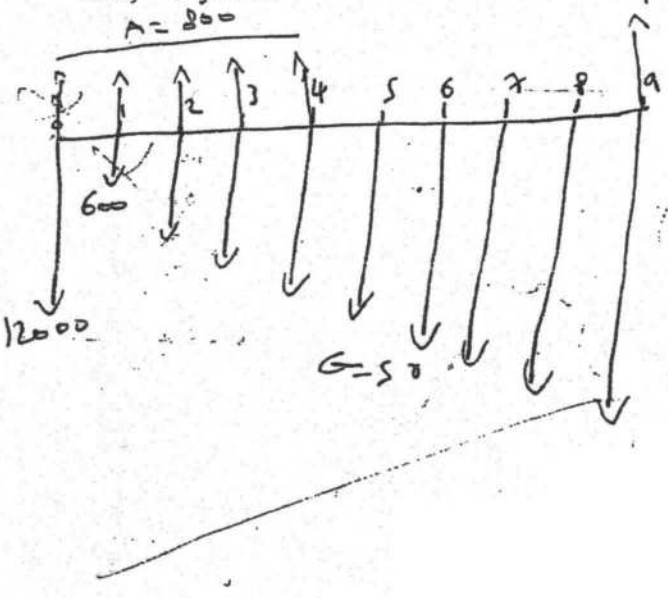
because $N * (P/F, i\%, 1)$

c) The change in the amount of money over a given time period is called the Time Value of Money. (T) F

Q 3: Draw the CFD for a project with the following payments:

- first cost = JD 12000;
- operating cost, started from the 2nd year = JD 600, increased by JD50;
- Annual Income for the first 5 years = JD 800;
- Salvage Value = 4100;

if n, = 9 years.



GOOD LUCK

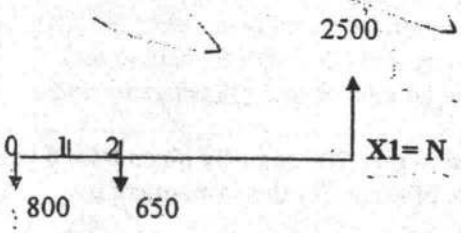
ENGINEERING ECONOMY FIRST EXAM. 16/11/09 ; MON.

NAME..... الإسم

SHOW YOUR CALCULATIONS.

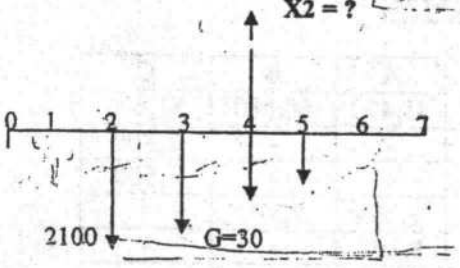
Q1: Find the Equivalent Values (X), (Use $i = 9\%$ per year).

X1= 7,1822 X2= 8635,1 X3= 702,99 X4= 878,029



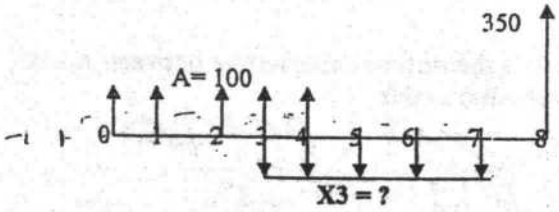
P = A(P/A,9%,2) P = F(P/F,9%,2) + 800 = 1397,077

[P = F(P/F,9%,n)

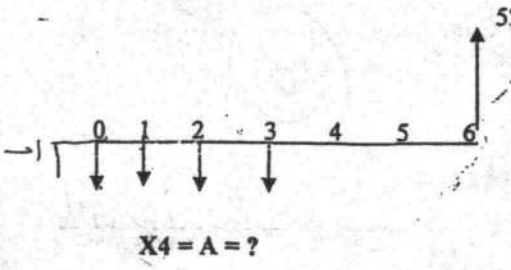


7 ← 0,54703
n ← 0,53884
8 ← 0,50187
n = 7,1822

P = A(P/A,9%,4) − G(P/G,9%,4) , F = P(F/P,9%,3)



P = A(P/A,1%,6) , F = P(F/P,9%,3) = 500,9



P = F(P/F,9%,18) = 208,69

A = P(A/P,9%,5)

= 202,99

P = F(P/F,9%,17) = 2844,556

A = P(A/P,9%,4) =

Engineering Economy second Exam 2012/2013 **B**

Student Name: الرقم الجامعي / الشعبة **Time: 50 min.**

Items	Alt.A	Alt.B	Alt.C	Alt.D	Alt.E
FC First Cost	23000	11000	5300	8500	14500
EC_2 Extra cost @end of 2nd yr	-	-	350	200	100
OC Operating Cost	-	1900	100	250	500
$M_{1-5}C$ Cost for the first 5yrs	2000	-	-	-	-
Extra Cost every 3years	-	250	-	-	250
Extra cost every 2 yrs	500	-	-	-	-
SV Salvage Value	-	1500	700	400	250
Life(n) yrs	∞	6	6	3	7
i%	15%	15%	15%	15%	15%

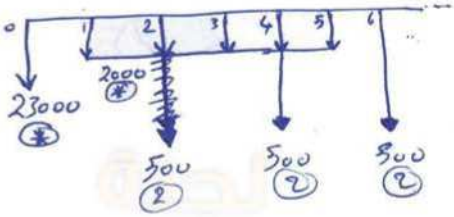
Q1: Using the capitalized cost criterion (PW) which is the optimal alternative A or B?

$i = 0.15$ $i = 15\%$

$C_1 = -23000$

$C_2 = -2000\,(P/A, 15\%, 5) = -6704.2$ (3.3521)

$C_3 = -[500\,(A/F, 15\%, 2)]/i = -1550.4$ (0.46512)



$C.C = +C_1 + C_2 + C_3 = -31254.6$

a- CC (A) is -31254.6

$C_1 = [-11000\,(A/P, 6)]/i = -19377.6$ (0.26424)

$C_2 = -1900/i = -12666.667$

$C_3 = [-250\,(A/F, 3)]/i = -479.9667$ (0.28798)

$C_4 = [1500\,(A/F, 6)]/i = 1142.4$ (0.11424)

$C.C = C_1 + C_2 + C_3 + C_4 = -31381.833$



b- CC(B) is -31381.833

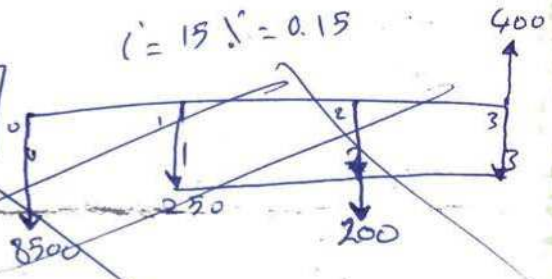
c- which is the optimal alternative Alt. (A)

* **Q2: Using the Present worth (PW) if we compare between C and D find the PW for D only ?**

The Present worth for alternative D is.... rejected

$$PW = -8500 - [250(P/A,3)] - [200(P/F,2)] + [400(P/F,3)]$$

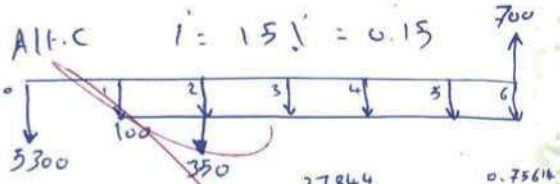
(2.2832, 0.75614, 0.65752)



$PW = -8959.02$ for (D) خلف الصفحة

$PW_D = -9523.832$

$PW_C = -5640.458$ for Alt c



$$P_W = -5300 - 100(P/A,6) - 350(P/F,2) + 700(P/F,6)$$

(3.7844, 0.75614, 0.43233)

$P_{W_C} = -5640.458$

Q3Using the Annual worth (AW) compare between A and E?

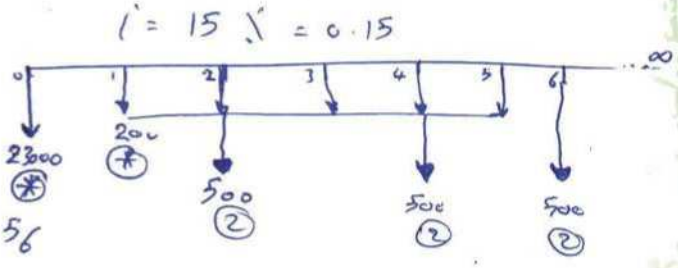
a- AW (a) $= -3783.123$

$A_1 = -23000 * i = -3450.0$

$A_2 = [-200(P/A,5)] * i = -100.563$ (3.3521)

$A_3 = -500(A/F,2) = A_{300} = -232.56$ (0.46512)

$A_W = A_1 + A_2 + A_3 = -3783.123$



b- AW (b) = ~~40~~

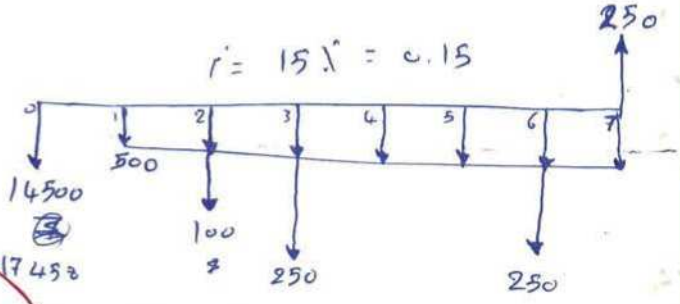
$A_1 = -14500(A/P,7) = -3485.22$ (0.24036)

$A_2 = -500$

$A_3 = -100(P/F,2)(A/P,7) = -18.17458$ (0.75614, 0.24036)

$A_4 = -250(A/F,3) = -71.995$ (0.28798)

$A_5 = 250(A/F,7) = 22.59$ (0.09036)



$A_{W_E} = -4052.7995$

c- which is the optimal alternative

Alt. (A) is optimal.

Q4: interest rate of 18 % per year compounded monthly find out,
Effective interest rate semiannually is... 0.09344263

18% y/m

$R_y = 0.18$

$R_{semi} = 0.18 \div 2 = 0.09$

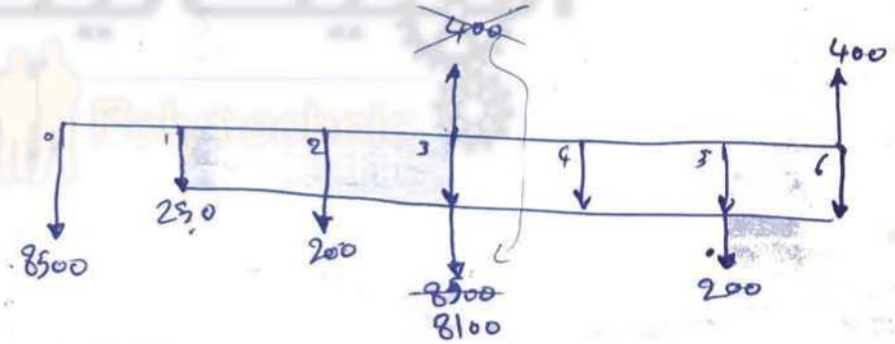
$m = 6$

$$i_{eff}^{semi} = \left(1 + \frac{R}{m}\right)^m - 1 = 0.09344263$$

Q2

L.C.M = 6

Alt. D:-



$$P_{W_D} = -8500 - [250(P/A, 6)] - 200(P/F, 2) - 200(P/F, 5) + 400(P/F, 6)$$

(3.7844, 0.75614, 0.49718, 0.43233)

$P_{W_D} = -9523.832$

Alt(D) Rejected.

NOMY B first exam second -term 2011/2012.

الشعبة ١٠-١١ NAME. الإسم ...

ALCULATIONS.

ivalent Values (X), (Use ***i.= 15%*** per year).

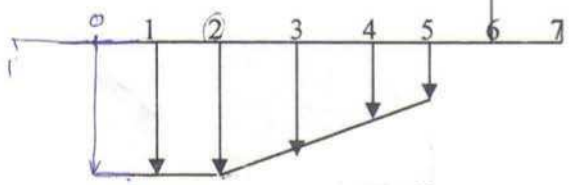
....X2= 2596.6 X3= 53.074 X4= 652.383

take for 50 to accumulate to 150 at an interest rate of 10% per year?

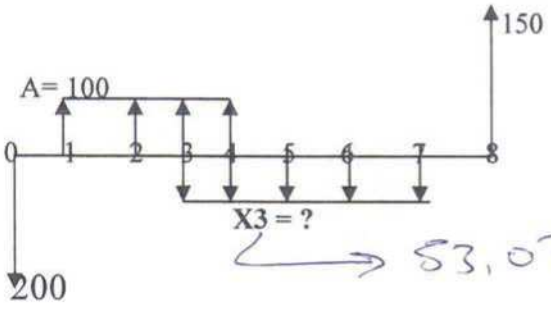
n= 11.53861

P= 50
F= 150
50 = 150 (P/F, 10%, n) = 1/3

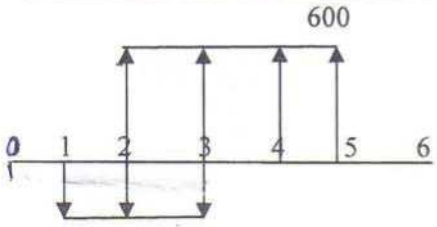
X2 = 2596.6



300 G=30



53,074



X4 = A = ? 652.383

ate of 10% per year, compounded monthly, the effective interest per six months(semiannually) is

i/semi= 5,105 %

Q2: Circle "T" (for True) or "F" (for False) for each of the following statements.

A) ECONOMICS The study of how limited resources is used to satisfy unlimited human wants
T F

B) To invest any amount for one year at a (8%) compound interest rate, is better than at a (6%) simple interest rate. T F

C) Risk: Possibility that the borrower will be unable to pay T F

D) INTEREST RATE the fee that a borrower pays to a lender for the use of his or her mone T F



Engineering Economy second Exam 2011

Student Name..... .. Time: 50 min.

Use the given data in the following Table to solve Q1 & Q2

Items	Alt.A	Alt.B	Alt.C	Alt.D
F.C	7000	12000	1800	3400
Extra cost @end of 3rd yr	2000	-	600	-
O.C	-	1000	90	250
A.C for the first 5yrs	800	-	-	-
Extra Cost every 3years	500	200	-	-
Extra cost every 5 yrs	-	-	-	-
A.I for the first 3yrs with increasing gradient(G=50)	600	-	200	-
S.V	-	2000	700	400
Life(n) yrs	∞	6	6	12
i%	10%	10%	12%	12%

Q1: Using the capitalized cost criterion which is the optimal alternative A or B?

a- CC1(A) is A = Pxi = 70000 / 0.10 = 70 000 ↓

b- CC2(A) is

c- CC3(A) is

d- CC4(A) is

e- CC5(A) is

f- CC (A) is .....

g- CC1(B) is

h- CC2(B) is

i- CC3(B) is

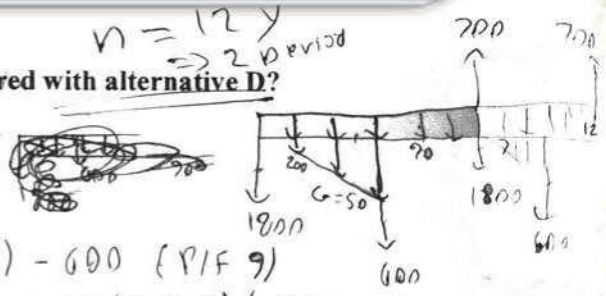
j- CC4 (B) is

k- CC5(B) is

l- CC(B) is .....

m- The decision is.....

Q2: find out the **Present worth** of alternative C which is compared with alternative D?



a-The Present worth for alternative C is.....

$-1800 - 900(P/A\ 12\%\ 12)$

$-1800(P/F\ 12\%\ 6) - 600(P/F\ 12\ 3) - 600(P/F\ 9)$

$-200(P/A\ 3) - 200(P/A\ 3)(P/F\ 6)$

$+700(P/F\ 6) + 700(P/F\ 12) - 50(P/G\ 3) - 50(P/G\ 3)(P/F\ 6)$

$= 458,831.8067$

b- find out the Annual worth of alternative D which is compared with alternative C?
the Annual worth of alternative D is.....

$A.w = -400(A/F\ 12\%\ 12) - 250$ (0.10842, 0.04144)

$A.w = -400(A/P\ 12\%\ 12) - 250 - 400(A/F\ 12\%\ 12)$

$= -939.204$

2

Q3: interest rate of 10 % yearly compounded **Continuous** find out,
A-Effective interest rate yearly is.....

$i_{eff} = e^{0.10} - 1 = 10.5171\%$

2

b- Effective interest rate yearly compounded quarterly is.....

$i_{eff.} = \left(1 + \frac{10.5171\%}{4}\right)^4 + 1 = 10.9391\%$

5

b- Effective interest rate semiannually compounded monthly is.....

$i.con.s = e^{0.05} - 1 = 0.051271$

$i.con.s/m = \left(1 + \frac{0.051271}{6}\right)^6 - 1 = 0.052379$

5

Good Luck

Engineering Economy second Exam 2011

Student Name..... **Time: 50 min.**

Use the given data in the following Table to solve Q1 & Q2

Items	Alt.A	Alt.B	Alt.C	Alt.D
F.C	7000	12000	1800	3400
Extra cost @end of 3rd yr	2000	-	600	-
O.C	-	1000	90	250
A.C for the first 5yrs "	800	-	-	-
Extra Cost every 3years	500	200	-	-
Extra cost every 5 yrs	-	-	-	-
A.I for the first 3yrs with increasing gradient(G=50)	600	-	200	-
S.V	-	2000	700	400
Life(n) yrs	∞	6	6	12
i%	10%	10%	12%	12%

Q1: Using the capitalized cost criterion which is the optimal alternative A or B?

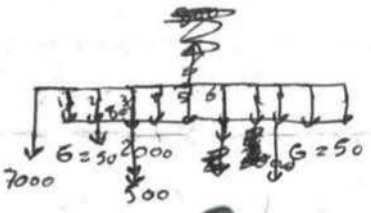
a- CC1(A) is -7000

b- CC2(A) is $-800(P/A\ 10\%,5)(P/A\ 10\%,20)$
$= -7946,15$

c- CC3(A) is $-500((P/F\ 10\%,3)+(P/F\ 10\%,6))$
$= -657,89$

d- CC4(A) is $-2000(P/F\ 10\%,3)$
$= -1502,62$

e- CC5(A) is
$-600 - 50((P/G\ 10\%,3)+(P/G\ 10\%,6))$
$= -600 - 600,658 = -1200,655$



f- CC (A) is -18309,315

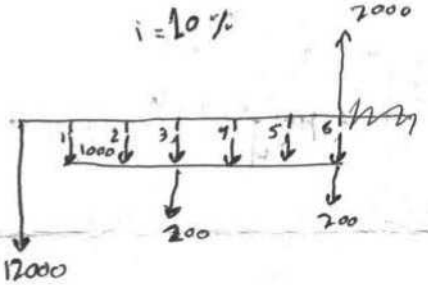
g- CC1(B) is -12000

h- CC2(B) is $-1000(P/A\ 10\%,6)$
$= -4355,2$

i- CC3(B) is $-200(P/F\ 10\%,3)$
$= -150,262$

j- CC4 (B) is $-200(P/F\ 10\%,6)$
$= -112,894$

k- CC5(B) is $2000(P/F\ 10\%,6)$
$= 1128,94$



l- CC(B) is $-12000 - 4355,2 - 150,262 - 112,894 + 1128,94 =$
$-15489,416$

m- The decision is B

Q2: find out the **Present worth** of alternative C which is compared with alternative D?

a-The Present worth for alternative C is.....

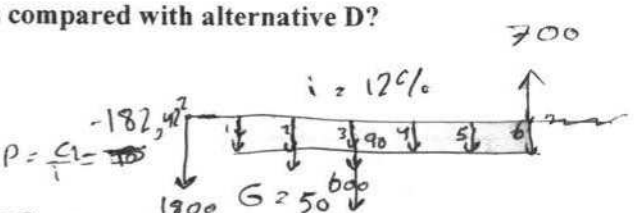
$C_1 = -1800$

$C_2 = -90\,(A/P\ 12\%\ 6) = -21,8907$

$C_3 = -200 - 50\,(A/G\ 12\%\ 3) = -246,23$

$C_4 = -600\,(P/F\ 12\%\ 3) = -427,068$

$C_5 = 700\,(P/F\ 12\%\ 6) = 354,641$



$P = \frac{C_1}{i} = -2051,916$

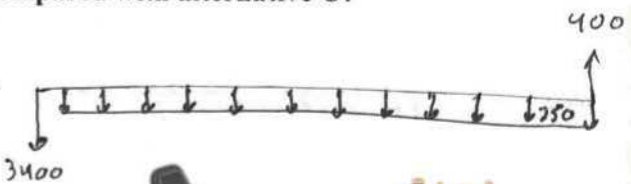
$P = \frac{C_4}{i} = -3558,9$

$P = \frac{C_5}{i} = 2955,341$

b- find out the Annual worth of alternative D which is compared with alternative C?
the Annual worth of alternative D is.....

$C_1 = -3400 \times 0,12 = -408$

$C_2 = 400\,(A/F\ 12\%\ 12) * 0,12 = 1,98912$



$A.W. = -408 - 250 + 1,98912$

$= -656,01$

Q3: interest rate of 10 % yearly compounded Continuous find out,
A-Effective interest rate yearly is.....10,517%

$I(eff) = e^r - 1$

$e^{0,1} - 1 = 10,517\ \%$

$I_{eff} = e^r - 1$

$= e^{0.10} - 1 = 10,517\%$

b- Effective interest rate yearly compounded quarterly is.....

$I_{(eff)/q} = \left(1 + \frac{r}{m}\right)^m - 1$

$= 1 + \left(\frac{10,517}{4}\right)^4 - 1 = 10,939\ \%$

y/q

$= \left(1 + \frac{r}{m}\right)^m - 1$

1 - e

b- Effective interest rate semiannually compounded monthly is.....

$I \div 2 = 10,517\% \div 2 = 0,0525$

$I_{(sem/mon)} = \left(1 + \frac{r}{m}\right)^m - 1 = \left(1 + \frac{0,0525}{6}\right)^6 - 1 = 5,366\%$

Good Luck

Engineering Economy Secord Exam Time: 60minutes Name: [redacted] ID: ٤٥٤ ٢-٢		Al-Balqa` Applied Universtiy Faculty of Engineering Technology Department of Civil Engineering

17/20

It is obligatory to show your all detailed calculations

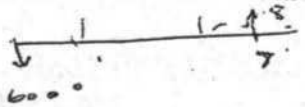
Part1:

1) If the effective interest rate per quarter is 5% compounded continuously then the effective interest rate per semiannual compounded monthly is.....10.1126 %.
2) If you deposit 1000 now, 600 two years later and 1200 after 6 years so the accumulated money at the end of year 7 at an interest rate 6% per semiannual compounded quarterly will be.....$.....
3) If 6000 is invested now which is expected to give 200 annually income for 8 years and 8000 at the end of year 8 thus the rate of return for this investment is.....6.26%

nominal

(3)

Part2:



Use i =12%

Cash item	A	B	C
F.C	45,423	66,580	51,600
Extra cost every 6 years	-	-	5,800
Annual cost starting @ end of yr3	-	-	1,500
A.O.C	3,267	2,510	-
A.I	8,198	-	-
S.V	12,214	11,290	-
Income @ the end of year 5	-	7,123	22,150
Project life (years)	8	6	∞

-Using AW criterion, decide which is the optimal alternative between A&B?
(Please show the C.F.D for each alternative)

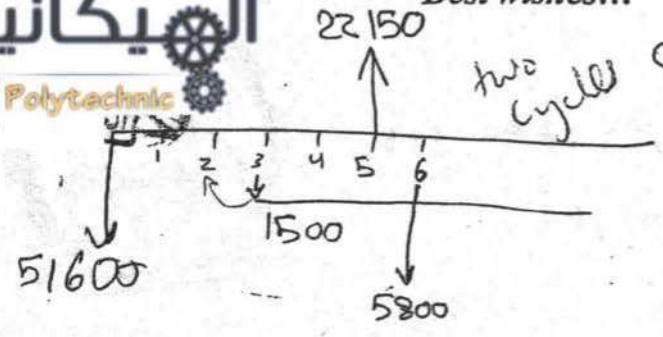
1) The AW for alternative A is.....-3219.6517.....
2) The AW for alternative B is.....-1619.1755.27.....
3) The decision is.....A.....

(1)

-The capitalized cost CC for alternative C is.....-56445.99

(6)

Best wishes...



$CC = -1500(P/F,12\%,2)0.12$
$-51600 - 5800(A/F,12\%,6)\frac{1}{0.12}$
$+ 22150(P/F,12\%,5)$

0.56743